

Literature Status: Problem 16 on Closed Ideals of Operators on Schreier Space

Run fa_banach_001 — lane 10

9 August 2026

Status: literature already answered. Problem 16 of arXiv:2006.15415 is answered by Theorem 4.4 of the separate paper arXiv:2008.12362. For every finite-order Schreier space $X[\mathcal{S}_N]$, $N \geq 1$, the lattice of closed operator ideals has the maximal possible cardinality 2^c .

1 Original question

Freeman, Schlumprecht, and Zsák ask in Problem 16, arXiv PDF page 18, of *Banach spaces for which the space of operators has 2^c closed ideals* [1]:

What is the cardinality of the lattice of closed ideals of the space of operators on Schreier space?

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Problem 16. *What is the cardinality of the lattice of closed ideals of the space of operators on Schreier space?*

Source Problem 16, arXiv PDF page 18.

2 Separate later answer

Manoussakis and Pelczar-Barwacz explicitly say in the introduction of *Small operator ideals on the Schlumprecht and Schreier spaces* that their finite-order Schreier result solves Problem 16 of the source paper [2]. Their Theorem 4.4, arXiv PDF page 14, states that for every $N \geq 1$ there is a family $(\mathcal{I}_A)_{A \subset \mathbb{R}}$ of small closed operator ideals on $X[\mathcal{S}_N]$ satisfying

$$\mathcal{I}_A \subset \mathcal{I}_B \iff A \subset B.$$

Thus there are 2^c distinct small closed ideals. This is also the upper bound for the total number of closed ideals on a separable Banach space, so the entire lattice has cardinality exactly 2^c .

Theorem 4.4. Fix $N \in \mathbb{N}$, $N \geq 1$. There is a family of small closed operator ideals $(I_A)_{A \subset \mathbb{R}}$ on the Schreier space $X[\mathcal{S}_N]$ such that $I_A \subset I_B \Leftrightarrow A \subset B$, for any $A, B \subset \mathbb{R}$. In particular there are exactly $2^{\mathfrak{c}}$ distinct small closed ideals on the Schlumprecht space. There is also a chain of cardinality $\mathfrak{c}$ of small closed ideals and an antichain of cardinality $2^{\mathfrak{c}}$ of small closed ideals on the Schreier space $X[\mathcal{S}_N]$.

Proof. Take a family $(T_{J_r})_{r \in \mathbb{R}} \subset \mathcal{S}(X_S)$ with $(J_r)_{r \in \mathbb{R}}$ as in Lemma 4.3 and for any $A \subset \mathbb{R}$ define I_A to be the closed operator ideal generated by $(T_{J_r})_{r \in A}$. Then obviously $A \subset B$ implies $I_A \subset I_B$. On the other hand, fix $A \subset \mathbb{R}$ and $r \in \mathbb{R} \setminus A$. Then by the choice of $(J_r)_{r \in \mathbb{R}}$ sets J_r and $(J_s)_{s \in A}$ satisfy the assumptions of Prop. 4.2 thus in particular $T_{J_r} \notin I_A$. In other words $T_{J_r} \in I_A$ implies $r \in A$, which finishes the proof. $\square$

Supporting Theorem 4.4, arXiv PDF page 14.

3 Scope and textual caveat

The result answers the classical Schreier-space case $X[\mathcal{S}_1]$ and, more strongly, every finite order $X[\mathcal{S}_N]$. It does not give a full lattice classification and does not cover infinite-order Schreier spaces.

The displayed theorem contains an evident one-word typo: after fixing ideals on the Schreier space $X[\mathcal{S}_N]$, one sentence says “on the Schlumprecht space.” The opening and final sentences of Theorem 4.4, its proof, the section heading, the abstract, and the introduction all identify the space as $X[\mathcal{S}_N]$. This packet therefore records the unambiguous Schreier-space reading and flags the typo explicitly.

This is an exact literature answer, not an original result. A bounded search through 9 August 2026 checked the run indexes, both arXiv ids, exact question wording, and later Schreier-space closed-ideal literature. The run contains a different packet using arXiv:2008.12362, but not this Problem 16 identification.

References

- [1] D. Freeman, Th. Schlumprecht, and A. Zsák, *Banach spaces for which the space of operators has $2^{\mathfrak{c}}$ closed ideals*, Forum Math. Sigma **9** (2021), e27; arXiv:2006.15415; DOI 10.1017/fms.2021.23.
- [2] A. Manoussakis and A. Pelczar-Barwacz, *Small operator ideals on the Schlumprecht and Schreier spaces*, J. Funct. Anal. **281** (2021), 109156; arXiv:2008.12362; DOI 10.1016/j.jfa.2021.109156.